IMPERIAL COLLEGE LONDON

EE4-25
EE9-CS1-2
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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Friday, 15 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E.C. Kerrigan

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1. Consider the feedback system in Figure 1. Here, $G(s)$ is a nominal system model and $K(s)$ is a controller. The transfer matrices $\Delta_a(s)$ and $\Delta_m(s)$ denote stable uncertainties.

The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable when:

- $\Delta_a = 0$ and $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|, \forall\omega$, and,
- $\Delta_m = 0$ and $\|\Delta_a(j\omega)\| \leq |w_a(j\omega)|, \forall\omega$,

where $w_a(s)$ and $w_m(s)$ are appropriate weighting functions. The performance specification is to design an internally stabilizing controller such that when $\Delta_a = 0$ and $\Delta_m = 0$, $\|y(j\omega)\| < |w(j\omega)^{-1}|\|r(j\omega)\|, \forall\omega$, where $w(s)$ is an appropriate weighting function.

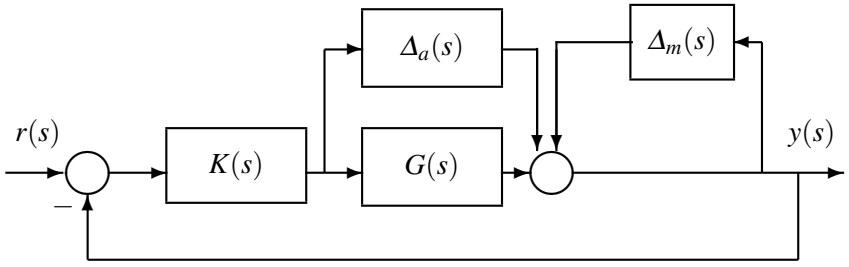


Figure 1

a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix Δ and a feedback transfer matrix H. [4]

b) Give an interpretation of the small gain theorem for single-input-single-output systems in terms of the Nyquist stability criterion. [4]

c) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_a(s)$, $w_m(s)$ and $w(s)$ that are sufficient to achieve the specifications. [4]

d) Define appropriate cost, external, control and measured signals and draw a block diagram, showing these signals, the nominal system model, the controller, and weighting functions. [4]

e) Derive and draw a generalised regulator problem that captures these sufficient conditions. [4]

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2. a) Consider a state–variable model described by the dynamic equation

$$\begin{bmatrix} \dot{x}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

and let $H(s) = D + C(sI - A)^{-1}B$ be the transfer matrix. Let

$$M = \begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix},$$

with $P = P^T$ and $\gamma > 0$.

i) State and prove LMI conditions involving A and P that guarantee that A is stable. [4]

ii) State and prove LMI conditions involving P and M that guarantee that A is stable and $\|H\|_\infty < \gamma$. [4]

iii) State dual equivalent conditions involving A and H for the stability of A and for $\|H\|_\infty < \gamma$. Hence state dual LMI sufficient conditions for the stability of A and for $\|H\|_\infty < \gamma$ involving P and another matrix N of a similar form to M. [4]

b) Consider the feedback system in Figure 2. Let $T_{yu}(s)$ denote the transfer matrix from u to y. An internally stabilizing gain matrix F is to be designed such that, for given $\gamma > 0$, $\|T_{yu}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{yu}(s)$. [4]

ii) Derive sufficient conditions for the existence of a feasible F. Your conditions should be in the form of the existence of solutions to linear matrix inequalities. [4]

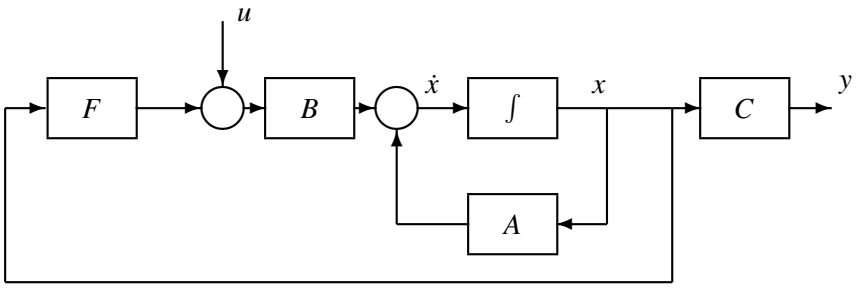


Figure 2

3. a) Suppose that the n-th order transfer matrix $G(s)$ has a state space realization

$$G(s) \stackrel{s}{=} \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right]$$

and assume that A is stable. Let

$$AP + PA^T + BB^T = 0, \qquad A^T Q + QA + C^T C = 0,$$

with

$$P = P^T = \begin{bmatrix} 0 & 0 \\ 0 & P_2 \end{bmatrix}, \qquad Q = Q^T = \begin{bmatrix} Q_1 & 0 \\ 0 & 0 \end{bmatrix},$$

where $\mathscr{R}^{n_1 \times n_1} \ni Q_1 \succ 0$ and $\mathscr{R}^{n_2 \times n_2} \ni P_2 \succ 0$ and where $n_1 + n_2 = n$. Prove that the realization $G(s)$ can be decomposed into two subsystems:

i) A subsystem with n_1 modes that are stable, uncontrollable and observable. [4]

ii) A subsystem with n_2 modes that are stable, controllable and unobservable. [4]

Draw a diagram showing the two subsystems of $G(s)$. [4]

b) Let

$$\mathscr{S} = \{(A_1, C_1), \ldots (A_N, C_N)\},$$

where $A_i \in \mathscr{R}^{n \times n}$ and $C_i \in \mathscr{R}^{m \times n}, i = 1, \ldots, N$ are given. The set $\mathscr{S}$ is called simultaneously detectable if there exists $L \in \mathscr{R}^{n \times m}$ such that $A_i + LC_i$ is stable for $i = 1, \ldots, N$.

i) Suggest necessary and sufficient conditions, in the form of matrix inequalities, for the simultaneous detectability of $\mathscr{S}$. [4]

ii) Suggest sufficient conditions, in the form of linear matrix inequalities, for the simultaneous detectability of $\mathscr{S}$. Explain how L may be obtained when these conditions are satisfied. [4]

4. Consider the $\mathscr{H}_\infty$ regulator shown in Figure 4 for which it is assumed that the triple (A,B,C) is minimal and $x(0)=x_0$. Let $H(s)$ denote the transfer matrix from w to $z=[\,z_1^T\ \ z_2^T\,]^T$ and let $\gamma>0$ be given.

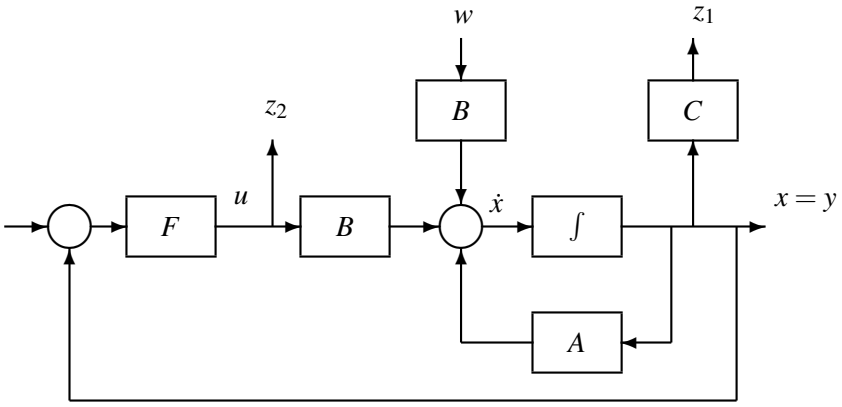


Figure 4

An internally stabilizing matrix F is to be designed such that $\|H\|_\infty \le \gamma$ for a given $\gamma>0$.

a) Derive the generalized regulator formulation for this design problem. [4]

b) By defining a Lyapunov function and a cost function J involving an auxiliary variable $X=X^T$, and carrying out completions of squares, derive an expression for the cost function J that can be used to solve the design problem. [4]

c) Use the expression for J above to formulate the problem and derive sufficient conditions for its solution. Your conditions should have the form of the existence of a certain solution to an algebraic Riccati equation. It should also include an expression for the gain matrix F as well as an expression for the worst–case disturbance. [4]

d) By using a limiting value of γ, use the solution above to derive the solution of the LQR problem. Write down the minimum value of the cost function, the Riccati equation and the optimal value of F. [4]

e) Use the Bounded Real Lemma to derive an LMI solution to the $\mathscr{H}_\infty$ regulator problem shown in Figure 4. [4]